

KhataAI Backend Engineering Blueprint

Version 1.0

1. Architecture

Pattern: Router → Service → Repository → MongoDB. Use FastAPI, Motor (MongoDB), Pydantic v2, JWT, Docker, Redis (later), ReportLab, Gemini, Whisper.

2. Folder Structure

app/ core/, db/, middleware/, exceptions/, dependencies/, routes/, services/, repositories/, schemas/, models/, utils/, ai/, background/, tests/, main.py

3. Core Modules

Settings, DatabaseManager, LoggingMiddleware, Security, JWT utilities, Exception handlers, Response helpers.

4. Repositories

UserRepository, StoreRepository, CustomerRepository, ProductRepository, TransactionRepository, ExpenseRepository, NotificationRepository, AuditLogRepository, VoiceTranscriptRepository.

5. Services

AuthService, StoreService, CustomerService, ProductService, InventoryService, TransactionService, LedgerService, ExpenseService, VoiceService, SpeechService, LLMService, ExtractionService, PDFService, NotificationService, AnalyticsService, AuditLogService.

6. Schemas

Auth, Store, Customer, Product, Transaction, Expense, Notification, AuditLog, VoiceTranscript, Common Responses.

7. APIs

/auth, /store, /customers, /products, /inventory, /transactions, /voice, /expenses, /analytics, /notifications, /audit-logs, /admin, /health

8. Database Collections

users, stores, customers, products, transactions, voice_transcripts, expenses, notifications, audit_logs

9. Business Rules

No transaction deletion. Use reversal transactions. Voice always requires preview before save. AI extracts only; backend calculates GST and totals. Soft delete customers/products using status/active flag. Every query filtered by store_id from JWT.

10. Security

JWT Access + Refresh tokens, bcrypt password hashing, role-based authorization, HTTPS, rate limiting, audit logs.

11. Production

Daily backups, MongoDB indexes, CloudWatch logging, Docker deployment, health endpoint, Redis caching (future).

12. Development Order

1. Core setup 2. Authentication 3. Store Profile 4. Customers 5. Products 6. Transactions & Ledger 7. Voice AI 8. PDF Invoice 9. Analytics 10. Notifications 11. Audit Logs 12. Deployment & Testing